Maharashtra Education Society's
Institute of Management and Career Courses (IMCC), Pune
AUTONOMOUS

[CGPA-3.32] in Cycle 1

Academic Year 2026-27
FY MCA- SEM I-DIV A
IT12 Data Structure and Algorithms
Practical Assignment1

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```
class Node:
  def __init__(self, value):
    self.data = value
    self.next = None

class LinkedList:
  def __init__(self):
    self.head = None

  # 1. create singly linked list-append(creates nodes with positive and negative
numbers)
  def create_list(self, values):
    for value in values:
      self.insert_at_end(value)

  def insert_at_end(self, data):
    new_node = Node(data)

    if self.head is None:
      self.head = new_node
    else:
      current_node = self.head

      while current_node.next is not None:
          current_node = current_node.next

      current_node.next = new_node

  # 2. Traverse and print the node values
  def display(self):
    current_node = self.head      #starting at first node using head

    while current_node is not None:     # traverse until last node
      print(current_node.data, end=" -> ")
      current_node = current_node.next
```

```
    print()

  # 3. Insert node at a specific position
  def insert_at_position(self, new, pos):
    if(pos == 1):
       new.next = self.head
       self.head = new
    else:
      target = 1
      temp = self.head

      while(target != pos - 1 and temp.next != None):
        temp = temp.next
        target += 1

      new.next = temp.next
      temp.next = new

  # 4. Delete the nodes having negative values
  def delete(self):
    temp = self.head
    prev = None

    while(temp != None):
      if(temp.data < 0):
        if(prev == None):
          self.head = temp.next
        else:
          prev.next = temp.next
      else:
        prev = temp

      temp = temp.next

# Main program
linked_list = LinkedList()

linked_list.create_list([10, -20, 30, -35, 40])

print("Original list:")
linked_list.display()

print("After inserting 25 at position 2:")
linked_list.insert_at_position(Node(25), 2)
linked_list.display()
```

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```
print("After deleting negative values from the list:")
linked_list.delete()
linked_list.display()
```

```
PS C:\Users\Tanvi\OneDrive\Documents> python .\dsaPrac.py
Original list:
10 -> -20 -> 30 -> -35 -> 40 ->
After inserting 25 at position 2:
10 -> 25 -> -20 -> 30 -> -35 -> 40 ->
After deleting negative values from the list:
10 -> 25 -> 30 -> 40 ->
```